

Software Test Plan (STP) - Voice Assistant for Multimedia Controls

Project: Voice Assistant for Multimedia Controls

Version: 1.0

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Date: 15-09-2026

Status: Draft for Stage 2 Submission

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1. Introduction

Purpose: This document defines the test plan for the Voice Assistant for Multimedia Controls v1.0. It outlines the test objectives, scope, strategy, resources, schedule, responsibilities, and verification approach.

Scope: Testing covers the voice-controlled multimedia functions defined in the SRS: microphone input, speech-to-text conversion, intent detection, command validation, media playback, track navigation, volume control, feedback, logging, error handling, performance, accuracy, reliability, privacy, and security. Gesture recognition, camera input, MediaPipe, and general-purpose voice-assistant functions are excluded.

References: Team 10 Software Requirements Specification (SRS) v1.0, Voice Assistant for Multimedia Controls; project design specifications; selected speech-recognition and VLC/media-player documentation used for implementation.

Definitions: STP (Software Test Plan), SRS (Software Requirements Specification), STT/ASR (Speech-to-Text / Automatic Speech Recognition), UI (User Interface), VLC (VideoLAN media player), FR (Functional Requirement), NFR (Non-Functional Requirement), RTM (Requirements Traceability Matrix), UAT (User Acceptance Testing).

2. Test Items

- USB microphone / audio input module
- Speech recognition module
- Intent detection and command-processing module
- Command validation module
- Media controller and VLC/media-player integration
- Volume control module
- Feedback and error-handling module
- Logging module
- Startup and recovery handling
- Security and privacy controls

3. Features to be Tested

Features mapped to SRS requirement IDs:

- VA-F-001: Capture voice input from the USB microphone
- VA-F-002: Convert captured speech into text
- VA-F-003: Detect the intent of a recognized multimedia command
- VA-F-004: Validate detected intents against the supported-command allow-list
- VA-F-005: Play media
- VA-F-006: Pause media
- VA-F-007: Resume media
- VA-F-008: Stop media
- VA-F-009: Select the next media item
- VA-F-010: Select the previous media item
- VA-F-011: Increase volume
- VA-F-012: Decrease volume
- VA-F-013: Mute audio
- VA-F-014: Recognize common equivalent command phrases
- VA-F-015: Provide successful command feedback
- VA-F-016: Handle unrecognized or unsupported commands
- VA-F-017: Record timestamped command, intent, and execution-result events
- VA-F-018: Recover gracefully from speech-recognition/media-player errors
- VA-NF-001: Command-to-action response time ≤ 3 seconds for at least 90% of recognized commands
- VA-NF-002: Intent classification accuracy $\geq 90\%$
- VA-NF-003: Two-hour continuous operation without application crash
- VA-NF-004: No unintended multimedia actions from invalid/low-confidence commands
- VA-NF-005: Understandable feedback within 2 seconds when enabled
- VA-NF-006: No raw microphone recordings retained during normal operation
- VA-NF-007: Application ready within 30 seconds of startup
- VA-SR-001 to VA-SR-006: Security and privacy controls

4. Features Not to be Tested

- Gesture recognition, camera input, and MediaPipe, because this release is voice-only.
- General-purpose conversational assistant behavior outside the defined multimedia command set.
- Internal implementation details of third-party speech-recognition services outside the team's control.
- Internal VLC source-code behavior; only the interfaces and multimedia functions used by the project are tested.
- Unrelated Raspberry Pi operating-system functions not required by the application.
- Raw-audio model training or dataset generation beyond the controlled accuracy tests defined by the SRS.

5. Test Approach / Strategy

Levels:

- Unit tests (module-level)
- Integration tests (microphone ↔ speech recognition ↔ command processing ↔ media control)
- System tests (end-to-end voice command workflow)
- Acceptance tests (UAT)

Types:

- Functional testing (core voice and multimedia features)
- Negative testing (unsupported, unclear, and low-confidence commands)
- Regression testing
- Performance testing (command response time)
- Accuracy testing (intent classification)
- Reliability and recovery testing
- Usability testing (feedback clarity and timing)
- Security and privacy testing

Entry Criteria: Stable build delivered, Raspberry Pi test environment ready, USB microphone operational, media player and test media available, test data prepared, and SRS-to-test mappings established.

Exit Criteria: 100% of planned test cases executed, 0 critical defects open, all high-priority SRS requirements verified, security validation completed, and all applicable acceptance criteria satisfied.

5.1 Security Validation

Security validation shall verify the security objectives and security requirements defined in SRS Section 5.1. Testing will be performed in a controlled local environment.

- Validate that only supported multimedia intents in the configured allow-list are executed (VA-SR-001).
- Validate that low-confidence or failed recognition is rejected before media execution (VA-SR-002).
- Verify that raw microphone audio is not stored during normal operation (VA-SR-003).
- Test recognized-text input against OS-command injection attempts and verify that arbitrary operating-system commands are never executed (VA-SR-004).
- Verify that command mappings and security configuration cannot be modified by unauthorized users (VA-SR-005).
- Inspect logs for unauthorized modification protection and confirm that passwords, tokens, and raw audio are not stored (VA-SR-006).
- Test unsupported, malformed, repeated, and unrelated phrases to ensure fail-safe behavior.

SRS Security ID	Validation Activity	Test Case
VA-SR-001	Allow-list validation for executable multimedia intents	TC-Sec-03
VA-SR-002	Confidence threshold / equivalent rejection rule	TC-Sec-04
VA-SR-003	Raw-audio storage inspection	TC-Privacy-01
VA-SR-004	OS-command injection resistance	TC-Sec-05
VA-SR-005	Configuration access-control validation	TC-Sec-06
VA-SR-006	Secure log content and permission audit	TC-Sec-07

6. Test Environment

Hardware: Raspberry Pi (Team 10 hardware), USB microphone, supported audio output device, microSD card, and power supply.

Software: Raspberry Pi OS/Linux, Python 3.x, configured speech-recognition engine/library, VLC or compatible media player, and project application build.

Tools: Git/GitHub, Python test framework, terminal/system-monitoring utilities, manual test logs, and screenshot capture tools.

Test Data: Controlled supported voice phrases, equivalent command phrases, unsupported phrases, low-confidence/noise cases, multimedia test files/playlists, and security test strings.

7. Test Schedule

Milestone	Schedule	Output
Test case design	15-09-2026	Test case document
Environment setup	After Test Case Design	Ready test environment
Test execution	After Environment Setup	Test execution results
Defect reporting & retesting	After Test Execution	Updated defect log / retest results
Final test summary	After Defect Reporting & Retesting	Test Summary Report
UAT / final demonstration	After Final Test Summary	UAT / demonstration evidence

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual and automated, where applicable)
- Test Scripts
- Test Data
- Test Execution Logs / Screenshots
- Defect Reports
- Test Summary Report
- Updated Requirements Traceability Matrix

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Keerthi B Patil	Prepare test cases, verify quality criteria, coordinate test execution, and record defects.
Test Engineer	Kushi Niranjana	Design and execute speech-recognition, intent-detection, and functional test cases.
Developer	Hemant Ram Nelapati	Support Raspberry Pi test setup, USB microphone/audio configuration, VLC integration testing, and defect fixing.
System Design & Integration Lead	Yash Raghavendra A	Define and review system requirements, oversee the overall system design, coordinate module integration, ensure consistency between requirements and implementation, and support final validation of the system.

10. Risks and Mitigation

Risk	Mitigation
Background noise affects speech recognition	Use a controlled testing environment, repeat test phrases, and document test conditions.
Speech-recognition engine depends on network, if applicable	Maintain stable connectivity and use an offline engine if supported by the final implementation.
Raspberry Pi resource limitations	Use lightweight components, monitor CPU/memory, and keep the test media set controlled.
VLC/media-player integration differences	Record and freeze the tested media-player version/configuration.
Microphone or driver failure	Perform a microphone smoke test before execution and maintain a known-working backup input path where possible.
Late software changes introduce regressions	Execute regression tests after changes to speech, intent, or media-control modules.
Team member availability delays testing	Share test artifacts through the common GitHub repository and distribute execution across active members.

11. Assumptions & Dependencies

- The Raspberry Pi hardware is available and stable during testing.
- The USB microphone is compatible with the Raspberry Pi configuration used by Team 10.
- The selected speech-recognition engine is available in the test environment.
- VLC or the selected media player supports the required playback and volume controls.
- A controlled set of multimedia files is available for repeatable tests.
- Testers are permitted to inspect local application files and logs for security/privacy validation.
- The Test Plan remains synchronized with the approved Team 10 SRS v1.0.

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable or unstable for an extended period.
- The build is too unstable and blocks more than 30% of planned test cases.
- A critical defect prevents safe execution of further testing.
- The microphone or media-player integration required for core tests is unavailable.
- A critical security or privacy issue makes continued normal testing unsafe.

Resume testing if:

- Blocking defects are resolved or an approved workaround is available.
- The test environment is restored and smoke-tested.
- A stable build is redeployed and baseline tests pass.
- Security/privacy issues are resolved or formally accepted for the next test phase.

13. Test Case Management & Traceability

RTM ensures mapping of SRS requirements to test cases.

The following traceability matrix maps the approved SRS requirement IDs to their corresponding test cases. All requirements are covered by at least one planned test case.

SRS Requirement ID	Requirement / Feature	Test Case(s)
VA-F-001	Microphone input capture	TC-Voice-01
VA-F-002	Speech-to-text conversion	TC-Voice-02
VA-F-003	Intent detection	TC-Intent-01
VA-F-004	Command allow-list validation	TC-Sec-03
VA-F-005	Play media	TC-Media-01
VA-F-006	Pause media	TC-Media-02
VA-F-007	Resume media	TC-Media-03
VA-F-008	Stop media	TC-Media-04
VA-F-009	Next media item	TC-Media-05
VA-F-010	Previous media item	TC-Media-06
VA-F-011	Volume up	TC-Vol-01
VA-F-012	Volume down	TC-Vol-02
VA-F-013	Mute audio	TC-Vol-03
VA-F-014	Equivalent phrase recognition	TC-Intent-02
VA-F-015	Success feedback	TC-FB-01
VA-F-016	Invalid/unrecognized handling	TC-Err-01
VA-F-017	Command/result logging	TC-Log-01
VA-F-018	Error recovery	TC-Recovery-01
VA-NF-001	Command response time	TC-Perf-01
VA-NF-002	Intent classification accuracy	TC-Acc-01
VA-NF-003	Two-hour continuous reliability	TC-Reliability-01
VA-NF-004	No unintended multimedia action	TC-Sec-03
VA-NF-005	Feedback usability/timing	TC-UX-01
VA-NF-006	Raw-audio privacy	TC-Privacy-01
VA-NF-007	Startup readiness	TC-Startup-01
VA-SR-001	Intent allow-list	TC-Sec-03
VA-SR-002	Confidence validation	TC-Sec-04
VA-SR-003	No raw-audio retention	TC-Privacy-01
VA-SR-004	Arbitrary OS-command prevention	TC-Sec-05
VA-SR-005	Configuration access control	TC-Sec-06
VA-SR-006	Secure logging	TC-Sec-07

13.1 Example Test Case Records

Test Case ID	Test Name	Procedure / Input	Expected Result	SRS Trace
TC-Voice-01	Microphone capture	Speak a supported command.	Voice input is captured without application error.	VA-F-001
TC-Voice-02	Speech recognition	Speak a known supported phrase.	Correct text is produced.	VA-F-002
TC-Intent-01	Intent detection	Speak one clear phrase for each core command.	Correct normalized intent is identified.	VA-F-003
TC-Intent-02	Equivalent phrases	Use at least three equivalent phrases for supported commands.	Equivalent phrases map to the same intent.	VA-F-014
TC-Media-01	Play	Say 'play' with media stopped.	Media starts playing.	VA-F-005
TC-Media-02	Pause	Say 'pause' during playback.	Playback pauses and position is retained.	VA-F-006
TC-Media-03	Resume	Say 'resume' after pause.	Playback resumes.	VA-F-007
TC-Media-04	Stop	Say 'stop' during playback.	Playback stops.	VA-F-008
TC-Media-05	Next	Say 'next' with a playlist.	Next available item is selected.	VA-F-009
TC-Media-06	Previous	Say 'previous' with playlist/history.	Previous available item is selected.	VA-F-010

Test Case ID	Test Name	Procedure / Input	Expected Result	SRS Trace
TC-Vol-01	Volume up	Say 'volume up'.	Volume increases without exceeding maximum.	VA-F-011
TC-Vol-02	Volume down	Say 'volume down'.	Volume decreases without going below minimum.	VA-F-012
TC-Vol-03	Mute	Say 'mute'.	Audio is muted.	VA-F-013
TC-FB-01	Feedback	Execute a valid command.	Success feedback is produced.	VA-F-015
TC-Err-01	Invalid input	Speak unsupported/unclear command.	Error feedback appears and no action is taken.	VA-F-016
TC-Log-01	Logging	Execute commands and inspect logs.	Timestamp, intent and result are recorded; raw audio is absent.	VA-F-017
TC-Recovery-01	Recovery	Simulate speech/media component failure.	Application handles error without crash where recovery is possible.	VA-F-018
TC-Perf-01	Response time	Measure command-to-action time across controlled runs.	At least 90% complete within 3 seconds.	VA-NF-001
TC-Acc-01	Intent accuracy	Run 100 supported command samples.	At least 90% are classified correctly.	VA-NF-002
TC-Reliability-01	Soak test	Run application continuously for 2 hours.	No unplanned application crash.	VA-NF-003
TC-Sec-03	Invalid/allow-list test	Use unsupported or unrelated recognized text.	No multimedia action occurs.	VA-F-004, VA-NF-004, VA-SR-001
TC-Sec-04	Confidence validation	Supply unclear/low-confidence speech cases.	Command is rejected before media action.	VA-SR-002
TC-Privacy-01	Raw-audio privacy	Run normal commands and inspect storage.	No raw audio recordings remain by default.	VA-NF-006, VA-SR-003
TC-Sec-05	OS command injection	Use recognized strings resembling OS commands.	No OS command is executed.	VA-SR-004
TC-Sec-06	Configuration access	Attempt protected configuration changes without authorization.	Unauthorized changes are blocked.	VA-SR-005
TC-Sec-07	Secure logging	Inspect logs and file permissions.	No secrets/raw audio; log protection is verified.	VA-SR-006
TC-UX-01	Feedback usability	Review feedback during command tests.	Feedback is understandable and within target timing.	VA-NF-005
TC-Startup-01	Startup readiness	Start application and measure ready time.	Ready state reached within 30 seconds in $\geq 90\%$ of runs.	VA-NF-007

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % test cases passed / failed / blocked
- Defect density
- Defect aging
- Requirement coverage
- Intent classification accuracy
- 90th-percentile command response time
- Number of critical/high security issues

Reports:

- Test execution status during active testing
- Defect reports with severity, owner, and resolution status
- Updated Requirements Traceability Matrix
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Keerthi B Patil	Keerthi 15-09-2026
Test Engineer	Kushi Niranjana	Kushi 15-09-2026
Developer	Hemant Ram Nelapati	Hemant 15-09-2026
System Design & Integration Lead	Yash Raghavendra A	Yash 15-09-2026